Convex Optimization Exercises - Sheet 3

Exercise 9. Perform the simplex method for the following linear program. Discuss, which entering/leaving variable is the best choice in the first pivot step.

$$\begin{aligned} \text{minimize} \quad & f(x,y,z) = 2x + 3y + 4z \\ \text{subject to} \quad & 3x + 2y + z \leq 10 \\ & 2x + 5y + 3z \leq 15 \\ & x, y, z \geq 0 \end{aligned}$$

Exercise 10. Perform the simplex method for the following linear program. Observe that there is no unique solution and find all of them.

$$\begin{aligned} \text{minimize} \quad & f(x,y) = x + y \\ \text{subject to} \quad & 3x + y \leq 6 \\ & x + y \leq 4 \\ & x \geq 0 \end{aligned}$$

Exercise 11. Recall the definition of the following norms on $\mathbb{R}^n$:

- $\|x\|_\infty = \max(|x_1|, |x_2|, \ldots, |x_n|)$
- $\|x\|_1 = |x_1| + |x_2| + \cdots + |x_n|$

Discuss the n-dimensional crosspolytope $\{x \in \mathbb{R}^n \mid \|x\|_1 \leq 1\}$. What are its vertices? Describe it as the intersection of 2^n halfspaces.

Exercise 12. The norm approximation problem for a system of linear equation $Ax = b$ is the problem of minimizing $\|Ax - b\|$, for a given norm $\|\cdot\|$. Show that both

(i) minimize $\|Ax - b\|_\infty$

(ii) minimize $\|Ax - b\|_1$

are equivalent to an LP. Note that in (b) we are looking for an LP that is of polynomial size.

Exercise 13. Let $P = \{x \in \mathbb{R}^n \mid a_i^T x \leq b_i,\ i = 1, \ldots, m\}$ be a polytope. The *Chebyshev center* of P is the center of the largest Euclidean ball $B = \{x_c + u \mid \|u\| \leq r\}$ that lies in P. Model the problem of finding the Chebyshev center as a Linear Program.